ELEMENTARY MATH TRIGONOMETRY FUNCTION

Ex. No.: 01

Date:

AIM:

To perform Basic Arithmetic Operations and plot the Trigonometric function Using Scilab

PROCEDURE:

Step 1: Start all programs and search Scilab software

Step 2: Create New Scilab file from the File Menu

Step 3: Type the program

Step 4: Compile the program and check output

Step 5: Save the program file and exit the application

PROGRAM:

For Arithmetic operations

```
a=input("a:");
b =input("b:");
c=a+b;
disp('The Sum of a and b')
disp(c)
d=a-b
disp('The Difference of a and b')
disp(d)
e=a*b;
disp('The product of a and b')
disp(e)
f=a/b;
disp('The division of a and b')
disp(f)
```

For plot the Trigonometric function

```
// Define your x-values
x = 0 : .1 : 10*%pi;
// Define your function
y = cos(x);
// To Plot a function
plot(x, y)
```

```
// Add title, labels and legend
title('cos(x)')
xlabel('x'); ylabel('y');
legend('y = cos(x)')
```

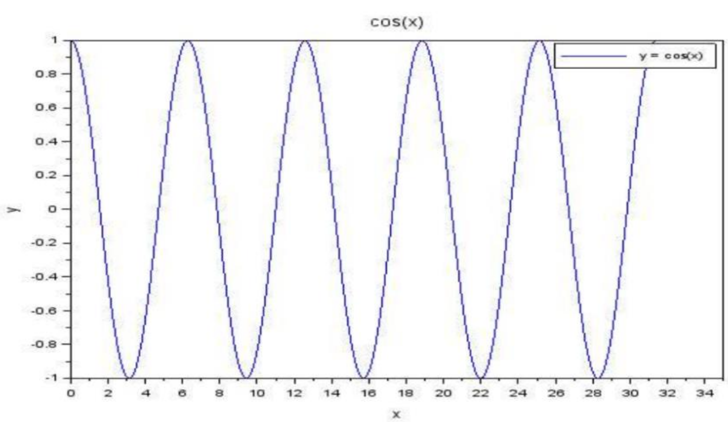
OUTPUT:

For Arithmetic operations

```
a:2
b:4
 "The Sum of a and b"
  6.
 "The Difference of a and b"
 -2.
 "The product of a and b"
  8.
 "The division of a and b"
  0.5
```

For plot the Trigonometric function

Plot:



Result:

Hence we perform Basic Arithmetic Operations and plot the Trigonometric function Using Scilab

LARGEST OF LIST OF NUMBERS

Ex. No.: 02

Date:

AIM

To find the Largest of given numbers

PROCEDURE:

Step 1: Start all programs and search Scilab software

Step 2: Create New Scilab file from the File Menu

Step 3: Type the program

Step 4: Compile the program and check output

Step 5: Save the program file and exit the application

PROGRAM:

```
a = input ("Enter the first number: ")
b = input ("Enter the second number: ")
c = input ("Enter the third number: ")
if(a>b && a>c) then
disp("a is the Largest number")
elseif(b>a && b>c) then
disp("b is the Largest number")
elseif(c>a && c>b) then
disp("c is the Largest number")
end
```

OUTPUT

```
Enter the first number: 10
Enter the second number: 15
Enter the third number: 5
b is the Largest number
```

RESULT

We have found out the largest of given numbers.

INPUT VECTOR

Ex. No.: 03

Date:

AIM:

To input the row and column vector

PROCEDURE:

Step 1: Start all programs and search Scilab software

Step 2: Create New Scilab file from the File Menu

Step 3: Type the program

Step 4: Compile the program and check output

Step 5: Save the program file and exit the application

PROGRAM:

```
clc
function []=Vector(m, A, B)
disp ('The Row vector is')
disp (A)
disp ('The column vector is')
disp (B)
endfunction
```

OUTPUT:

```
Vector (3,[1 2 3],[4 ;7 ;3])
 "The Row vector is"
  1.   2.   3.
 "The column vector is"
  4.
  7.
  3.
```

RESULT:

Hence the input vector is found

USER INPUT MATRIX

Ex. No.: 04

Date:

AIM

To finding the determinant and inverse of the given matrix.

PROCEDURE:

Step 1: Start all programs and search Scilab software

Step 2: Create New Scilab file from the File Menu

Step 3: Type the program

Step 4: Compile the program and check output

Step 5: Save the program file and exit the application

PROGRAM:

```
clc;
m=input('Enter the number of row ');
n=input('Enter the number of column ');
for i=1:m
for j=1:n
a(i,j)=input('enter the elements ');
end
end
A=matrix(a,m,n);
disp(A)
```

OUTPUT

```
Enter the number of row 2
Enter the number of column 2
enter the elements 4
enter the elements 6
enter the elements 5
enter the elements 7
  4.   6.
  5.   7.
```

RESULT

The matrix has been successfully inserted.

MATRIX MULTIPLICATION

Find the multiplication of matrix A=$\begin{bmatrix}1 & 2 & 3\\4 & 5 & 6\\7 & 8 & 9\end{bmatrix}$& B =$\begin{bmatrix}4 & 5 & 6\\7 & 8 & 9\\1 & 2 & 3\end{bmatrix}$

Ex. No.: 05

Date:

AIM:

To Multiply the given two matrices.

PROCEDURE:

Step 1: Start all programs and search Scilab software

Step 2: Create New Scilab file from the File Menu

Step 3: Type the program

Step 4: Compile the program and check output

Step 5: Save the program file and exit the application

PROGRAM:

```
clc
function []=Multiplication(m,n,A,B)
C=zeros (m,n);
C=A*B;
disp (“The first matrix is”)
disp (A)
disp (“The second matrix is”)
disp (B)
disp (“The product of two matrices is”)
disp (C)
endfunction
```

OUTPUT:

```
Multiplication (3,3,[1 2 3;4 5 6 ;7 8 9],[4 5 6 ;7 8 9 ;1 2 3])
“ The first matrix is”
 1.   2.   3.
 4.   5.   6.
 7.   8.   9.
“The second  matrix is”
4.  5. 6.
7.  8.  9.
```

1. 2. 3.

"The Product of two matrices is"

21.   27.   33.

57.   72.   87.

93.   117.   141.

RESULT:

Hence the multiplication of matrix is found

FINDING THE DETERMINANT, INVERSE OF A MATRIX

Ex. No.: 06

Date:

AIM

To finding the determinant and inverse of the given matrix.

PROCEDURE:

Step 1: Start all programs and search Scilab software

Step 2: Create New Scilab file from the File Menu

Step 3: Type the program

Step 4: Compile the program and check output

Step 5: Save the program file and exit the application

PROGRAM:

```
clc;
A=[1 2 ;4 5];
A=det(A);
disp(A,'The matrix is:');
disp(A,'The determinant of the given matrix is:');
A=[1 2 3;4 5 6;7 8 9];
A=det(A);
disp(A,'The matrix is:');
disp(A,'The determinant of the given matrix is:');
A=[6 7;8 9];
A=inv(A);
disp(A,'The matrix is:');
disp(A,'The inverse of the given matrix is:');
A=[9 8 7;6 5 4;3 2 1];
A=inv(A);
disp(A,'The matrix is:');
disp(A,'The inverse of the given matrix is:');
```

OUTPUT

```
  1.   2.
  4.   5.
 "The matrix is:"
  -3.
```

"The determinant of the given matrix is:"

```
 1.   2.   3.
 4.   5.   6.
 7.   8.   9.
```

"The matrix is:"

6.661D-16

"The determinant of the given matrix is:"

```
 6.   7.
 8.   9.
```

"The matrix is:"

```
-4.5   3.
 4.   -3.
```

"The inverse of the given matrix is:"

```
 9.   8.   7.
 6.   5.   4.
 3.   2.   1.
```

"The matrix is:"

```
-4.504D+15   9.007D+15  -4.504D+15
 9.007D+15  -1.801D+16   9.007D+15
-4.504D+15   9.007D+15  -4.504D+15
```

"The inverse of the given matrix is:"

RESULT

The determinant and inverse of a matrix of a given matrix has been successfully executed.

TRANSPOSE OF A MATRIX

Find the transpose of matrix A=$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$

Ex. No.: 07

Date:

AIM:

To find the transpose of matrix by using scilab.

PROCEDURE:

Step 1: Start all programs and search Scilab software

Step 2: Create New Scilab file from the File Menu

Step 3: Type the program

Step 4: Compile the program and check output

Step 5: Save the program file and exit the application

PROGRAM:

```
clc
A=input("Enter the matrix A")
disp(A)
[m,n]=size(A)
for i=1:n
for j=1:m
B(i,j)=A(j,i)
end
end
disp("Transpose is",B)
```

OUTPUT:

```
Enter the matrix A[1 2 3;4 5 6;7 8 9]
 1.   2.   3.
 4.   5.   6.
 7.   8.   9.
  "Transpose is"
   1.   4.   7.
   2.   5.   8.
   3.   6.   9.
```

RESULT: Hence the transpose of the matrix is found.

BISECTION METHOD

Find the real root of the equation$x^3 - x - 1 = 0$,correct to four decimal places by using Bisection method.

Ex. No.: 08

Date:

AIM:

To find the root of the equation for Bisection Method by using Scilab.

PROCEDURE:

Step 1: Start all programs and search Scilab software

Step 2: Create New Scilab file from the File Menu

Step 3: Type the program

Step 4: Compile the program and check output

Step 5: Save the program file and exit the application

ALGORITHM:

For any continuous function $f(x)$,

Step 1: Find the two points, say $a$and b, such that $a < b$ and $f(a) * f(b) < 0$

Step 2: Find the midpoint of a and b, say "c"

Step 3: "c" is the root of the given function if $f(c) = 0$; elsefollow the next step

Step 4: Divide the interval $[a, b]$

-If $f(c) * f(a) < 0$, there exist a root between c and a

-else if $f(c) * f(b) < 0$, there exist a root between c and b

Step 5: Repeat above three step until $f(c) = 0$

PROGRAM:

```
clc;clear;
disp("Bisection method")
deff('y=f(x)','y=x^3-x-1')
a=1
b=2
d=0.00001
i=0
while abs(a-b)>d
c=(a+b)/2
if f(c)*f(a)>0
```

```
a=c
else
b=c
i=i+1
disp([i,c])
end
end
```

OUTPUT:

Bisection method

1.	1.5
2.	1.375
3.	1.34375
4.	1.328125
5.	1.3261719
6.	1.3251953
7.	1.3249512
8.	1.3248291
9.	1.3247681
10.	1.3247375
11.	1.3247223

RESULT:

Hence the root of the equation is found.

REGULA FALSI METHOD

Find the roots of equation $x^3 - 100 = 0$, correct to four decimal places by using regula Falsi Method.

Ex. No.: 09

Date:

AIM:

To type a Regula Falsi Method by using Scilab.

PROCEDURE:

Step 1: Start all programs and search Scilab software

Step 2: Create New Scilab file from the File Menu

Step 3: Type the program

Step 4: Compile the program and check output

Step 5: Save the program file and exit the application

PROGRAM:

```
deff('d=f(x)','d=x^3-100')
a=input("Enter the value of a:")
b=input("Enter the value of b:")
n=input("Enter the number of iterations n:")
for i=1:n
   c=(a*f(b)-b*f(a))/(f(b)-f(a))
   disp([i,c])
if f(a)*f(c)<0 then
   b=c
end
if f(b)*f(c)<0 then
   a=c
end
c1=(a*f(b)-b*f(a))/(f(b)-f(a))
if abs(c1-c)<0.00001 then
   disp("We get accurate roots")
   break;
   end
end
```

OUTPUT:

Enter the value of a : 4

Enter the value of b : 5

Enter the value of n : 5

1. 4.5901639
2. 4.6377884
3. 4.6413098
4. 4.6415684
5. 4.6415873

We get accurate roots

RESULT:

Hence the root of the equation is 4.6415

Newton Raphson Method

Find the real root of the equation $x^3 - 100 = 0$, correct to four decimal places by using Newton Raphson Method.

Ex. No.: 10

Date:

AIM:

To find the root of the equation by Newton Raphson Method using Scilab.

PROCEDURE:

Step 1: Start all programs and search Scilab software

Step 2: Create New Scilab file from the File Menu

Step 3: Type the program

Step 4: Compile the program and check output

Step 5: Save the program file and exit the application

ALGORITHM:

1. Define a function $f(x) = 0$ as required using scilab.
2. Define the derivative function $f'(x) = 0$.
3. If two points are given in which the root lies then calculate the average of both, if they are not given then choose a suitable starting value according to the function.
4. Initialize the counter variable i=1.
5. Use the formula of Newton Raphson method inside the loop. We are using x to store the current value as well as next approximation as we are using loop.
6. Loop up to desired iterations and stop.
7. Print the value of root and 'We get required accuracy'.

Newton Raphson formula is $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$

PROGRAM:

```
disp("Newton Raphson Method")
deff('y=f(x)','y=x^3-100')
deff('y=z(x)','y=3*x^2')
a=input("Enter value of interval a:")
b=input("Enter value of interval b:")
```

```
n=input("Enter the number of iteration n:")
x0=(a+b)/2
for i=1:n
disp([i,x0])
x1=x0-f(x0)/z(x0)
if abs(x1-x0)<0.00001 then
disp("We get required accuracy")
break;
end
x0=x1
end
```

OUTPUT:

"Newton Raphson Method"

Enter value of interval a: 1

Enter value of interval b: 2

Enter the number of iteration n: 8

1. 1.5
2. 15.814815
3. 10.676485
4. 7.4100871
5. 5.5471188
6. 4.7813669
7. 4.6456353
8. 4.6415924

"We get required accuracy"

RESULT:

Hence the root of the equation is obtained.